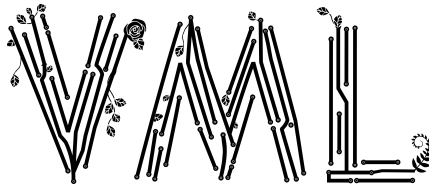


STEREOSCOPIC LAYER SHIFTER

By the Vibrant Media Lab,
University of Pittsburgh



Introduction

Stereoscopic Layer Shifter is a plugin for Inkscape, a free, open source vector graphics program. This plugin adds stereoscopic support to Inkscape. With this software, any image or document that is organized into a series of layers, composed of vectors, bitmapped graphics, or both, can be instantly converted into a stereoscopic 3D image. The final 3D image can be rendered in SBS or as separate L and R images.

It is particularly useful for creating:

- Stereoscopic titles for 3D video and VR productions. This enables you to edit SBS footage in nearly any video editor, importing titles from Inkscape.
- Stereoscopic slide shows. Simple or complex slides can be created in Inkscape and then exported in 3D for projection or screen display.
- Stereoscopic print media

Installation

You must first have Inkscape installed on your computer. Inkscape can be freely downloaded here: <https://inkscape.org/release/>

To install Stereoscopic Layer Shifter:

1. Open Inkscape.
2. Open the Preferences from the Edit menu.
3. Select the "System" section.
4. Scroll to the entry, "Inkscape Plugins"
5. Note the location (folder). Open that folder through your OS.
6. Locate your downloaded plugin, unzip it if you haven't already, and locate these two files:
stereo_extension.inx
stereo_extension.py
7. Copy these two files to the directory you opened in step 5.
8. Restart Inkscape.

How to Use

The key to creating a successful 3D final image is to properly organize your project.

First, set the aspect ratio correctly via File -> Document Properties. You can use any units you wish, but set up the aspect ratio properly. For slide shows, that's usually 4:3, so you could enter a width of 400mm and height of 300mm. For video titles (3D or VR180), check your actual final SBS video file, then size your document to *half* of the video width and the full height, converting pixels into mm. For print media, enter the size of the final desired print.

Next, create a series of layers. (In Inkscape, click the Layers pane, then the + (plus) button.) Label one "window". This will be the media depth: the depth of the screen or printed material that the image will be displayed upon. Any images or text that you create on this layer will appear to be "on" the display media. This is the "stereo window."

Any images or text that you place on layers above the window layer will appear to lift off the surface and float in the air (negative parallax). The higher the layer, the closer to the viewer the contents will float.

Any images or text you place on layers below the window layer will appear to recede behind the surface. The lower the layer, the farther back its contents will recede.

Your layers do not need to be symmetrical, with the window layer in the middle. You can have more layers in front, or more layers behind, or even *all* of your layers behind or in front of the window layer. But you must always have a window layer!

Next, populate your layers with the text and images you wish. You can create new vector images for these layers, or import layers from another program. For example, if you use Photoshop or a specialized drawing application to create digital art, make sure you compose in layers there, then export each layer separately. In Inkscape, place each image imported into the proper layer.

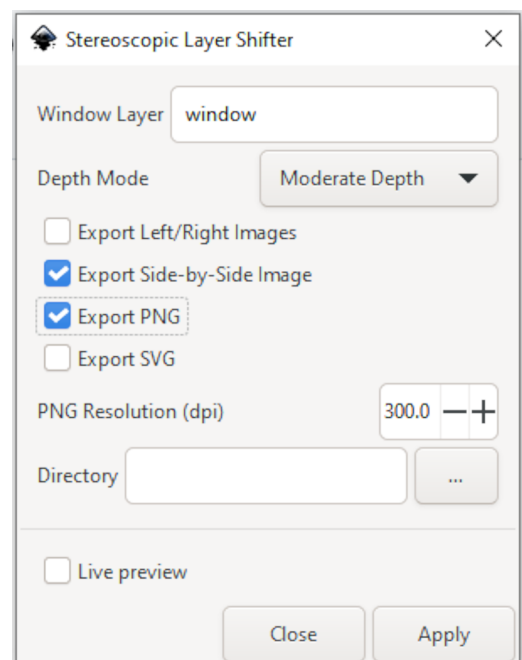
Exporting

Once you've completed your "stack" of elements and are ready to export them in 3D, go to the Extensions menu -> 3D -> Stereoscopic Layer Shifter

A window will pop up.

In "Window Layer" type the name of your window layer (usually "window").

Select between 3 different Depth Modes. "Moderate Depth" will produce a pleasing stereo image that works in most cases (recommended). "Conservative Depth" will produce a stereo image designed to minimize eyestrain. "Maximum Depth" will produce the most dramatic



stereoscopic depth, but should be tested to make sure it isn't difficult to fuse when enlarged into its final form.

In the check boxes, choose to export separate L/R images, a single SBS image, or both (you must click at least one of these).

You can also choose to export a rasterized PNG image (recommended most of the time), or to preserve any vector graphics by exporting an SVG, or both.

If you are exporting a rasterized PNG image, enter the resolution here. 300 will be correct for print media. Slideshows will likely utilize a smaller dpi. For video, experiment and see what resolution your final image ends up, then go back and adjust this for lower or higher resolution.

In "Directory," browse to your desired output folder.

Do NOT click "Live Preview." This standard Inkscape function does not work with this plugin and will cause it to crash.

Click "Apply" to create your output render.

For video work, you may need to resize your final output to match your video resolution if the output is much too large or won't import correctly in your video editing software.

Credits

Stereoscopic Layer Shifter was coded by Dharma Swaroop as part of the Vibrant Media Lab's "Immersive Media Initiative" at the University of Pittsburgh. The Vibrant Media Lab is directed by Zachary Horton. www.vml.pitt.edu

Open Source

Stereoscopic Layer Shifter is released as open source software under the GPL3 license. The source code can be accessed here: <https://github.com/Vibrant-Media-Lab/stereoVision>